Problem 1
Represent the derivative of the following scalar functions with respect to $\mathbf{X} \in \mathbb{R}^{D \times D}$.

(a) $f(\mathbf{X}) = tr(\mathbf{X}^2)$. Here, $tr(A)$ is the trace of a square matrix A.

(b) $g(\mathbf{X}) = tr(\mathbf{X}^3)$.

(c) $h(\mathbf{X}) = tr(\mathbf{X}^k)$ for $k \in \mathbb{N}$.

Solution:

(a)

$$tr(\mathbf{X}^2) = \sum_{j=1}^{D} \sum_{i=1}^{D} x_{ji} x_{ij}, \tag{1}$$

with $[\mathbf{X}]_{ij} = x_{ij}$. So, the derivative

$$\begin{aligned} \frac{d\, tr(\mathbf{X}^2)}{dx_{pq}} &= \frac{d}{dx_{pq}} \left(\sum_{i=1}^{D} \sum_{j=1}^{D} x_{ij} x_{ji} \right) && (2) \\ &= 2x_{qp}, && (3) \end{aligned}$$

for $p, q = 1, 2, \ldots, D$. Thus, we get the answer

$$\frac{df(\mathbf{X})}{d\mathbf{X}} = 2\mathbf{X}^\top. \tag{4}$$

(b) When a similar derivation is applied, we get

$$\begin{aligned} \frac{d\, tr(\mathbf{X}^3)}{dx_{pq}} &= \frac{d}{dx_{pq}} \left(\sum_{i=1}^{D} \sum_{j=1}^{D} \sum_{k=1}^{D} x_{ij} x_{jk} x_{ki} \right) && (5) \\ &= 3 \sum_{l=1}^{D} x_{ql} x_{lp}, && (6) \end{aligned}$$

for $p, q = 1, 2, \ldots, D$. So, the answer is

$$\frac{dg(\mathbf{X})}{d\mathbf{X}} = 3(\mathbf{X}^2)^\top. \tag{7}$$

(c) We can extend the derivation to general $k \in \mathbb{N}$:

$$\frac{d\, tr(\mathbf{X}^k)}{dx_{pq}} = k \sum_{i_1=1}^{D} \cdots \sum_{i_k=1}^{D} x_{qi_1} \cdots x_{i_k p}, \tag{8}$$

for $p, q = 1, 2, \ldots, D$. So, the answer is

$$\frac{dh(\mathbf{X})}{d\mathbf{X}} = k(\mathbf{X}^{k-1})^{\top}. \tag{9}$$

Problem 2

In order to alleviate overfitting in logistic regression, regularization technique can be used with the L_2-norm of the weight parameter, $||\mathbf{w}||^2 = \sum_{i=1}^{D} w_i^2$. When we are given $\mathbf{x}_1, \ldots, \mathbf{x}_N \in \mathbb{R}^D$ and $y_1, \ldots, y_N \in \{0, 1\}$ for training set, we want to derive the update rule for $\mathbf{w} \in \mathbb{R}^D$ in order to minimize the following loss function $L(\mathbf{w})$ having L_2-norm,

$$L(\mathbf{w}) = \sum_{i=1}^{N} \Big(-y_i \ln f(\mathbf{x}_i; \mathbf{w}) - (1 - y_i) \ln(1 - f(\mathbf{x}_i; \mathbf{w}))\Big) + ||\mathbf{w}||^2. \tag{10}$$

$$f(\mathbf{x}; \mathbf{w}) = \frac{1}{1 + \exp(-\mathbf{w}^\top \mathbf{x})} \tag{11}$$

(1) Derive the update rule for $\mathbf{w}$ when we use gradient descent.

(2) Discuss the effect of L_2-norm regularization.

Solution:

(1) We try to obtain the following update rule:

$$\mathbf{w}_{t+1} = \mathbf{w}_t - \eta \frac{\partial L}{\partial \mathbf{w}}. \tag{12}$$

with small η for gradient descent.

If we take the derivative of the loss function,

$$\frac{\partial L}{\partial \mathbf{w}} = -\sum_{i=1}^{N} \left(\frac{y_i \frac{\partial f}{\partial \mathbf{w}}}{f(\mathbf{x}_i; \mathbf{w})} + \frac{(1 - y_i)(-\frac{\partial f}{\partial \mathbf{w}})}{1 - f(\mathbf{x}_i; \mathbf{w})} \right) + 2\mathbf{w}, \tag{13}$$

and we use $\frac{\partial f}{\partial \mathbf{w}} = f(1 - f)\mathbf{x}$ to obtain the derivative,

$$\begin{aligned}
\frac{\partial L}{\partial \mathbf{w}} &= -\sum_{i=1}^{N} \left(\frac{\mathbf{y}_i f(\mathbf{x}_i; \mathbf{w})(1 - f(\mathbf{x}_i; \mathbf{w}))\mathbf{x}_i}{f(\mathbf{x}_i; \mathbf{w})} + \frac{(1 - \mathbf{y}_i)(-f(\mathbf{x}_i; \mathbf{w})(1 - f(\mathbf{x}_i; \mathbf{w}))\mathbf{x}_i)}{1 - f(\mathbf{x}_i; \mathbf{w})} \right) + 2\mathbf{w} \\
&= -\sum_{i=1}^{N} (\mathbf{y}_i(1 - f(\mathbf{x}_i; \mathbf{w}))\mathbf{x}_i - (1 - \mathbf{y}_i) f(\mathbf{x}_i; \mathbf{w})\mathbf{x}_i) + 2\mathbf{w} \\
&= -\sum_{i=1}^{N} ((\mathbf{y}_i - \mathbf{y}_i f(\mathbf{x}_i; \mathbf{w}))\mathbf{x} + (-f(\mathbf{x}_i; \mathbf{w}) + \mathbf{y}_i f(\mathbf{x}_i; \mathbf{w}))\mathbf{x}_i) + 2\mathbf{w} \\
&= -\sum_{i=1}^{N} (\mathbf{y}_i - f(\mathbf{x}_i; \mathbf{w}))\mathbf{x}_i + 2\mathbf{w}.
\end{aligned} \tag{14}$$

Finally, we obtain the equation for the update

$$\mathbf{w}_{t+1} = (1 - 2\eta)\mathbf{w}_t + \eta \sum_{i=1}^{N} (\mathbf{y}_i - f(\mathbf{x}_i; \mathbf{w_t}))\mathbf{x}_i. \tag{15}$$

(2) Current $\mathbf{w}_t$ is decayed by the amount of $1 - 2\eta$ for every update. The update arrives at the equilibrium when the vector $\frac{1}{2}\sum_{i=1}^{N}(\mathbf{y}_i - f(\mathbf{x}_i; \mathbf{w_t}))\mathbf{x}_i$ is equal to the current $\mathbf{w}_t$.

Problem 3

Consider the following function,

$$f(\mathbf{x};\mathbf{w}) = \frac{1-\exp\left(-\mathbf{w}^T\mathbf{x}\right)}{1+\exp\left(-\mathbf{w}^T\mathbf{x}\right)}, \tag{16}$$

having the shape in Fig. 1.

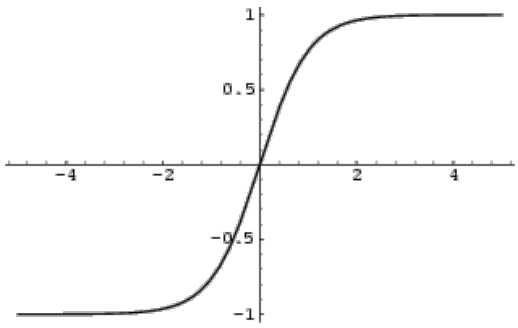

Figure 1: $f(\mathbf{x};\mathbf{w})$

Find the gradient descent update rule for $\mathbf{w}$ to minimize the loss

$$L = \frac{1}{2}\sum_{i=1}^{N}(f(\mathbf{x}_i;\mathbf{w}) - y_i)^2. \tag{17}$$

Here, we use N number of $\mathbf{x}_i \in \mathbb{R}^D$ and $y_i \in \{-1, 1\}$ for $i \in \{1, \ldots, N\}$ which are given in advance.

Solution:

We let

$$f(\mathbf{x};\mathbf{w}) = \frac{1-\exp\left(-g(\mathbf{x};\mathbf{w})\right)}{1+\exp\left(-g(\mathbf{x};\mathbf{w})\right)}. \tag{18}$$

with

$$g(\mathbf{x};\mathbf{w}) = \sum_{d=1}^{D} x_d w_d. \tag{19}$$

By using chain rule, the gradient $\frac{dL}{d\mathbf{w}}$ can be represented as

$$\frac{dL}{d\mathbf{w}} = \frac{\partial L}{\partial f}\cdot\frac{\partial f}{\partial g}\cdot\frac{\partial g}{\partial \mathbf{w}_t}. \tag{20}$$

Using the following calculation for each term in the gradient,

$$\frac{\partial L}{\partial f} = \sum_{i=1}^{N}(f(\mathbf{x}_i;\mathbf{w}) - y_i), \tag{21}$$

$$\frac{\partial f}{\partial g} = \frac{1}{2}(1 - f(\mathbf{x}_i;\mathbf{w}))(1 + f(\mathbf{x}_i;\mathbf{w})), \quad \text{and} \tag{22}$$

$$\frac{\partial g}{\partial \mathbf{w}_t} = \mathbf{x}_i, \tag{23}$$

we obtain the update rule for gradient descent:

$$\mathbf{w}_{t+1} = \mathbf{w}_t - \eta \cdot \frac{1}{2} \sum_{i=1}^{N} (f(\mathbf{x}_i; \mathbf{w}_t) - y_i)(1 - f(\mathbf{x}_i; \mathbf{w}))(1 + f(\mathbf{x}_i; \mathbf{w}))\mathbf{x}_i. \tag{24}$$

Problem 4

For a given data set $\mathcal{D} = \{\mathbf{x}_i, y_i\}_{i=1}^N$, derive the closed-form solution for $\mathbf{w}$ that maximizes the following probability $\boldsymbol{P}$.

$$\boldsymbol{P} = \prod_{i=1}^{N} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(y_i - f(\mathbf{x}_i; \mathbf{w}))^2\right), \tag{25}$$

with $f(\mathbf{x}; \mathbf{w}) = \mathbf{w}^\top \mathbf{x}$ for $\mathbf{x}, \mathbf{w} \in \mathbb{R}^D$.

Solution:

We consider the log of $\boldsymbol{P}$,

$$\ln \boldsymbol{P} = \sum_{i=1}^{n} -\frac{1}{2}\ln(2\pi\sigma^2) - \frac{1}{2\sigma^2}(y_i - f(\mathbf{x}_i; \mathbf{w}))^2, \tag{26}$$

and its derivative with respect to $\mathbf{w}$,

$$\begin{aligned}
\frac{d \ln \boldsymbol{P}}{d\mathbf{w}} &= \frac{d}{d\mathbf{w}}\left[\sum_{i=1}^{n} -\frac{1}{2}\ln(2\pi\sigma^2) - \frac{1}{2\sigma^2}(y_i - f(\mathbf{x}_i; \mathbf{w}))^2\right] \\
&= \sum_{i=1}^{n} -\frac{1}{2\sigma^2} \cdot 2(y_i - f(\mathbf{x}_i; \mathbf{w})) \cdot -\frac{d}{d\mathbf{w}} f(\mathbf{x}_i; \mathbf{w}) \\
&= \sum_{i=1}^{n} \frac{1}{\sigma^2}(y_i - \mathbf{w}^T\mathbf{x}_i)\mathbf{x}_i. \qquad (\because \frac{d}{d\mathbf{w}} f(\mathbf{x}_i; \mathbf{w}) = \frac{d}{d\mathbf{w}}\mathbf{w}^T\mathbf{x}_i = \mathbf{x}_i)
\end{aligned} \tag{27}$$

Now we obtain the optimal $\mathbf{w}^*$ using the condition $\frac{d \ln P}{d\mathbf{w}}\Big|_{\mathbf{w}=\mathbf{w}^*} = 0$.

$$\begin{aligned}
&\frac{d \ln P}{d\mathbf{w}}\Big|_{\mathbf{w}=\mathbf{w}^*} = 0 \\
&\Leftrightarrow \sum_{i=1}^{n} \frac{1}{\sigma^2}(y_i - \mathbf{w}^T\mathbf{x}_i)\mathbf{x}_i = 0 \\
&\Leftrightarrow \sum_{i=1}^{n} \mathbf{x}_i\mathbf{x}_i^T\mathbf{w} - \sum_{i=1}^{n} y_i\mathbf{x}_i = 0 \\
&\Leftrightarrow \mathbf{X}\mathbf{X}^T\mathbf{w} - \mathbf{X}\mathbf{y} = 0
\end{aligned} \tag{28}$$

From the final equation in Eq. (28), we obtain the following closed-form solution:

$$\mathbf{w}^* = (\mathbf{X}\mathbf{X}^T)^{-1}\mathbf{X}\mathbf{y}.$$

Problem 5

The Frobenius dot product $\langle \mathbf{A}, \mathbf{B} \rangle$ is defined as

$$\langle \mathbf{A}, \mathbf{B} \rangle = tr(\mathbf{A}^\top \mathbf{B}), \tag{29}$$

for $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{N \times N}$ using the scalar function $tr(\cdot)$ for the trace of a matrix,

$$tr(\mathbf{M}) = \sum_i^N M_{ii}, \quad \text{for } \mathbf{M} \in \mathbb{R}^{N \times N}. \tag{30}$$

Find the derivative of $\langle \mathbf{A}, \mathbf{B} \rangle$ with respect to $\mathbf{A}$ using the definition of the matrix derivative:

$$\left[\frac{d}{d\mathbf{A}} \langle \mathbf{A}, \mathbf{B} \rangle \right]_{ij} = \frac{\partial}{\partial A_{ij}} \langle \mathbf{A}, \mathbf{B} \rangle. \tag{31}$$

Solution:

The Frobenius dot product of $\mathbf{A}, \mathbf{B}$ is

$$\begin{aligned} \langle \mathbf{A}, \mathbf{B} \rangle &= tr(\mathbf{A}^\top \mathbf{B}) && (32) \\ &= \sum_{i=1}^N \sum_{j=1}^N A_{ij} B_{ij}. && (33) \end{aligned}$$

Then,

$$\begin{aligned} \left[\frac{d}{d\mathbf{A}} \langle \mathbf{A}, \mathbf{B} \rangle \right]_{kl} &= \frac{\partial}{\partial A_{kl}} \langle \mathbf{A}, \mathbf{B} \rangle && (34) \\ &= \frac{\partial}{\partial A_{kl}} \sum_{i=1}^N \sum_{j=1}^N A_{ij} B_{ij} && (35) \\ &= B_{kl}, && (36) \end{aligned}$$

for $k, l = 1, 2, \ldots, N$. Thus, the solution is

$$\frac{d}{d\mathbf{A}} \langle \mathbf{A}, \mathbf{B} \rangle = \mathbf{B}. \tag{37}$$

Problem 6

We are given a scalar function $f(\mathbf{X}) = \mathbf{w}^\top \mathbf{X}\mathbf{w}$ for $\mathbf{w} \in \mathbb{R}^D, \mathbf{X} \in \mathbb{R}^{D\times D}$. Find the derivative of $f(\mathbf{X})$ with respect to the vector $\mathbf{w}$. Use the definition of the vector derivative $\left[\frac{df}{d\mathbf{w}}\right]_k = \frac{\partial f}{\partial w_k}$, where w_k is the k-th element of $\mathbf{w}$.

Solution:

The function $f(\mathbf{X})$ can be rewritten as

$$
\begin{aligned}
f(X) &= \mathbf{w}^\top \mathbf{X}\mathbf{w} && (38)\\
&= \begin{pmatrix} w_1 & w_2 & \cdots & w_D \end{pmatrix} \begin{pmatrix} \mathbf{X}_{11} & \mathbf{X}_{12} & \cdots & \mathbf{X}_{1D} \\ \mathbf{X}_{21} & \mathbf{X}_{22} & \cdots & \mathbf{X}_{2D} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{X}_{D1} & \mathbf{X}_{D2} & \cdots & \mathbf{X}_{DD} \end{pmatrix} \begin{pmatrix} w_1 \\ w_2 \\ \vdots \\ w_D \end{pmatrix} && (39)\\
&= w_1\mathbf{X}_{11}w_1 + \cdots + w_D\mathbf{X}_{D1}w_1 + \cdots + w_1\mathbf{X}_{1D}w_D + \cdots + w_D\mathbf{X}_{DD}w_D && (40)\\
&= \sum_{j=1}^{D}\sum_{i=1}^{D} w_i\mathbf{X}_{ij}w_j \in \mathbb{R}. && (41)
\end{aligned}
$$

The derivative of $f(\mathbf{X})$ with respect to the w_k is calculated as

$$
\begin{aligned}
\frac{\partial f}{\partial w_k} &= \sum_{i=1}^{D} w_i\mathbf{X}_{ik} + \sum_{j=1}^{D} \mathbf{X}_{kj}w_j && (42)\\
&= \sum_{i=1}^{D} w_i(\mathbf{X}_{ik} + \mathbf{X}_{ki}), && (43)
\end{aligned}
$$

which leads to the solution:

$$
\begin{aligned}
\frac{df}{d\mathbf{w}} &= \begin{pmatrix} \sum_{i=1}^{D} w_i(\mathbf{X}_{i1} + \mathbf{X}_{1i}) \\ \sum_{i=1}^{D} w_i(\mathbf{X}_{i2} + \mathbf{X}_{2i}) \\ \vdots \\ \sum_{i=1}^{D} w_i(\mathbf{X}_{iD} + \mathbf{X}_{Di}) \end{pmatrix} && (44)\\
&= (\mathbf{X} + \mathbf{X}^\top)\mathbf{w} \quad \in \mathbb{R}^D. && (45)
\end{aligned}
$$